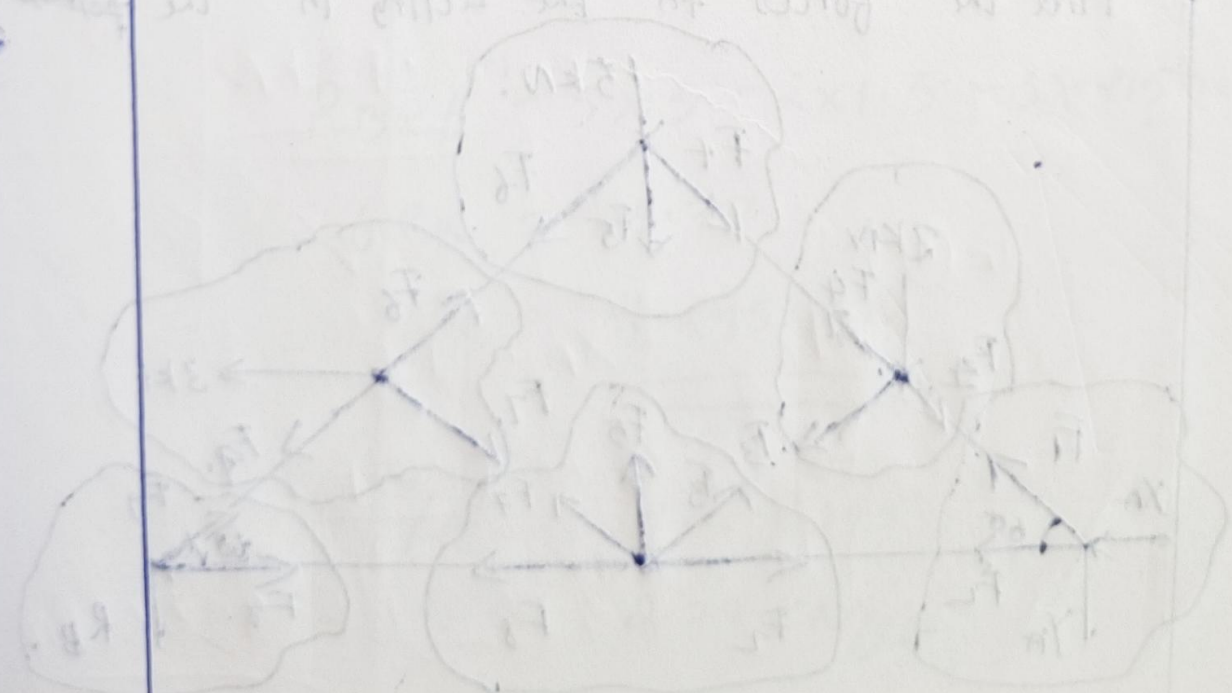
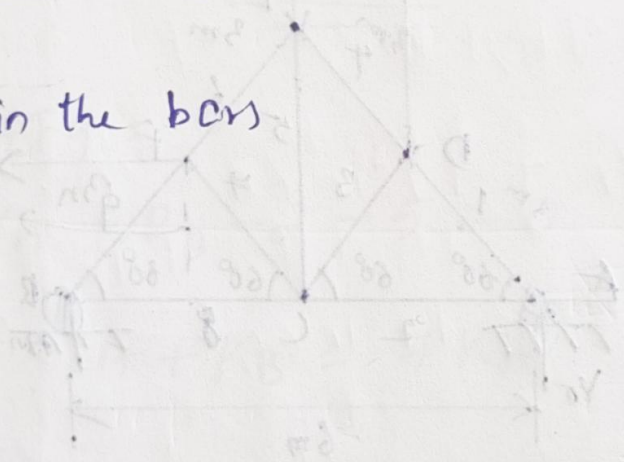


$n = 2j - 3 \rightarrow$ If this condition is satisfied, then the truss will be a perfect truss.

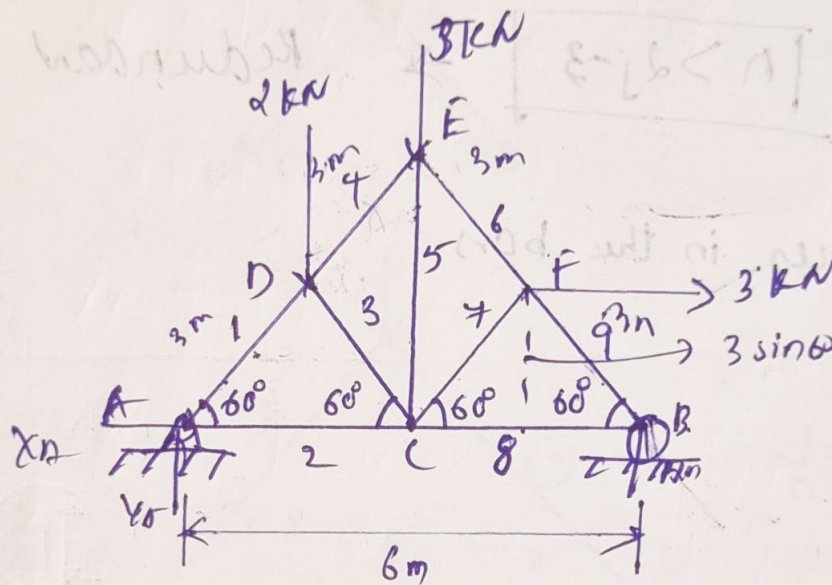
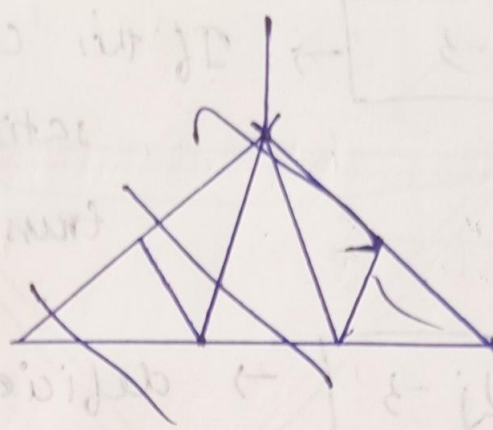
$n < 2j - 3 \rightarrow$ deficient truss.

$n > 2j - 3 \rightarrow$ Redundant truss.

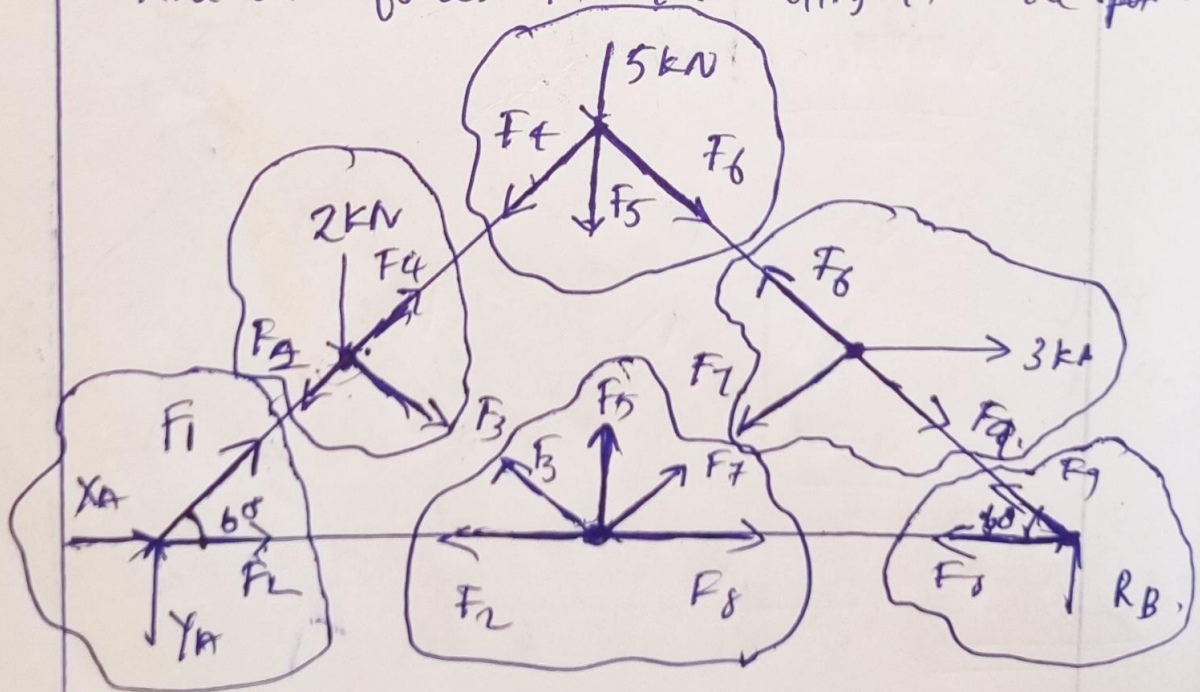
forces in the bars



22/4/24
Monday



Find the forces in the ~~the~~ acting in the ~~points~~ ^{joints}



$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum F_x = 0$$

$$X_A + 3 = 0$$

$$X_A = \underline{\underline{-3 \text{ kN}}}$$

$$\sum F_y = 0$$

$$Y_A + R_B - 2 - 5 = 0$$

$$Y_A + R_B = 7 \text{ kN}$$

$$\sum M_A = 0$$

$$R_B \times 6 - 5 \times 3 - 2 \times 1.5 - 3 \times 0.5 = 0$$

$$6 R_B = 25.5$$

$$R_B = \underline{\underline{4.25 \text{ kN}}}$$

$$Y_A = 7 - 4.25$$

$$= 2.75 \text{ kN}$$

~~ADP~~

Equilibrium about A

$$\sum F_H = 0$$

$$X_A + F_A + F_1 \cos 60^\circ = 0$$

$$X_A + F_A + \frac{F_1}{2} = 0 \rightarrow \textcircled{1}$$

$$\sum F_Y = 0$$

$$Y_A + F_1 \sin 60^\circ = 0$$

$$F_1 \sin 60^\circ = -2.71$$

$$F_1 = \frac{-2.71}{\sin 60^\circ}$$

$$= -3.13 \text{ kN}$$

In (b)

$$F_A = -X_A - \frac{F_1}{2}$$

$$= 3 - \left(\frac{-3.13}{2} \right)$$

$$= 3 + \frac{3.13}{2} = 4.56 \text{ kN}$$

Equilibrium about D

$$\sum F_x = 0$$

$$F_3 \cos 60^\circ - F_1 \cos 60^\circ = 0$$

$$F_3 \cos 60^\circ =$$

$$F_3 \cos 60^\circ + F_4 \cos 60^\circ - F_1 \cos 60^\circ = 0$$

$$F_3 + F_4 = F_1$$

$$F_3 + F_4 = \underline{\underline{-3 \cdot 1 \rightarrow (1)}}$$

$$\sum F_y = 0$$

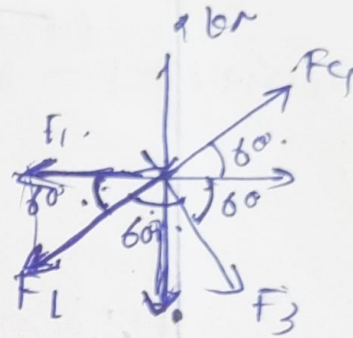
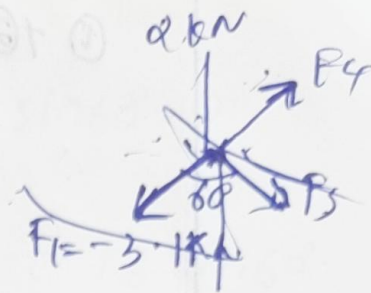
$$-2 - F_1 \sin 60^\circ - F_3 \sin 60^\circ + F_4 \sin 60^\circ = 0$$

$$F_4 \sin 60^\circ = 2 + F_1 \sin 60^\circ + F_3 \sin 60^\circ$$

$$F_4 = \frac{2}{\sin 60^\circ} + F_1 + F_3$$

$$F_4 - F_3 = 2 \cdot 30 + 3 \cdot 1$$

$$F_4 - F_3 = \underline{\underline{-0.8 \rightarrow (2)}}$$



① + ②

$$F_4 + F_3 = -3.1$$

$$F_4 - F_3 = -0.8$$

$$2F_4 = -3.9$$

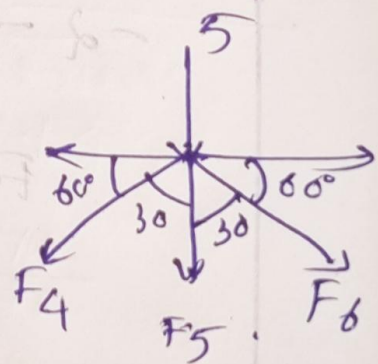
$$F_4 = -\cancel{2.95 \text{ kN}} // \underline{\underline{-1.98 \text{ kN}}}$$

$$F_4 + F_3 = -3.1$$

$$F_3 = -3.1 + 1.98$$

$$= \underline{\underline{-1.12 \text{ kN}}}$$

equilibrium about E



$$\sum F_x = 0$$

$$F_6 \cos 60^\circ - F_4 \cos 60^\circ = 0$$

$$\boxed{F_6 = F_4}$$

$$F_6 = \underline{\underline{-1.98 \text{ kN}}}$$

$$\sum F_y = 0.$$

$$-5 - F_5 - F_4 \sin 60^\circ - F_6 \sin 60^\circ = 0.$$

$$\begin{aligned} \rightarrow F_5 &= -5 + F_4 \sin 60^\circ + F_6 \sin 60^\circ \\ &= -5 - 1.98 \sin 60^\circ - 1.98 \sin 60^\circ \end{aligned}$$

$$F_5 = \underline{\underline{-1.57 \text{ kN}}}$$

About i^o

$$\sum F_x = 0$$

$$- F_7 \cos 60^\circ$$

$$+ F_9 \cos 60^\circ + 3 \text{ kN}$$

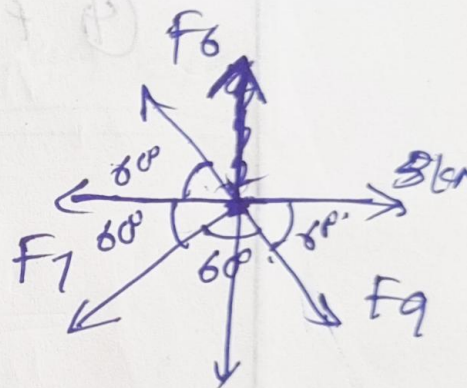
$$- F_6 \cos 60^\circ = 0.$$

$$- F_7 \cos 60^\circ + F_9 \cos 60^\circ = -3 + F_6 \cos 60^\circ$$

$$(F_9 - F_7) \cos 60^\circ = -3.99$$

$$F_9 - F_7 = -7.98$$

$$\underline{\underline{\hspace{2cm}}} \rightarrow \textcircled{3}$$



$$\sum F_y = 0$$

$$F_6 \sin 60^\circ - F_7 \sin 60^\circ - F_9 \sin 60^\circ = 0$$

$$(F_7 + F_9) \sin 60^\circ = F_6 \sin 60^\circ$$

$$F_7 + F_9 = F_6$$

$$F_7 + F_9 = 1.98 \rightarrow \textcircled{4}$$

$$\textcircled{4} + \textcircled{3}$$

$$F_9 - F_7 = -7.98$$

$$F_9 + F_7 = 1.98$$

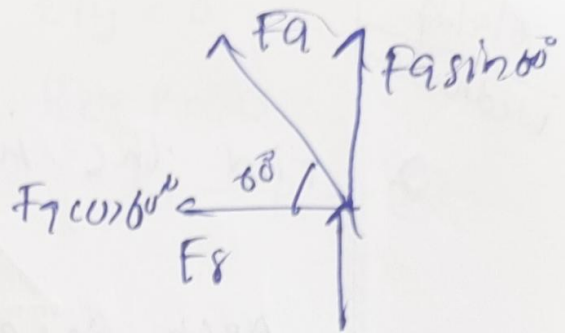
$$2F_9 = -9.96$$

$$F_9 = -4.98 \text{ kN}$$

$$F_7 = 1.98 + 4.98$$

$$= \underline{\underline{3 \text{ kN}}}$$

About B



$$\sum F_x = 0$$

$$- F_g \cos 60^\circ - F_8 = 0$$

$$F_8 = - F_g \cos 60^\circ$$
$$= - 4.98 \times \frac{1}{2}$$

$$= \underline{\underline{2.49 \text{ kN}}}$$

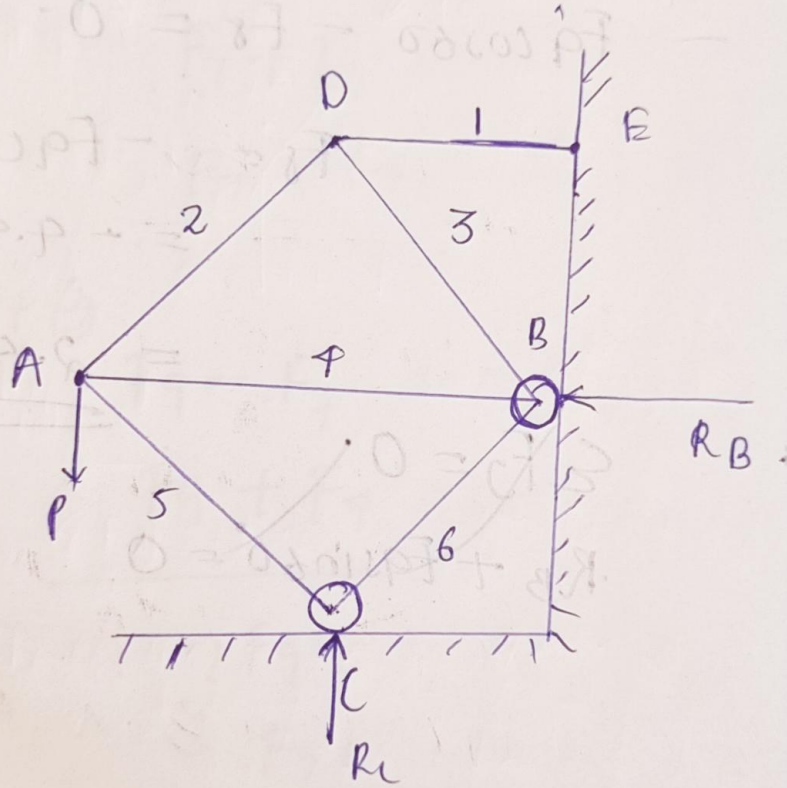
$$\sum F_y = 0$$

$$R_B + F_g \sin 60^\circ = 0$$

25/9/24
Wednesday

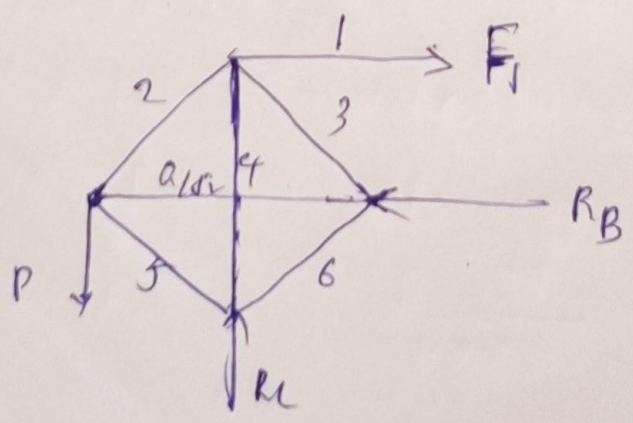
Q. Find the reaction forces

ABCD is a square with ^{diagonal} AB horizontal



Ans.

Free body diagram of ~~system~~ the truss.



$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$F_1 - R_B = 0$$

$$R_C - P = 0$$

$$F_1 = R_B \rightarrow (1)$$

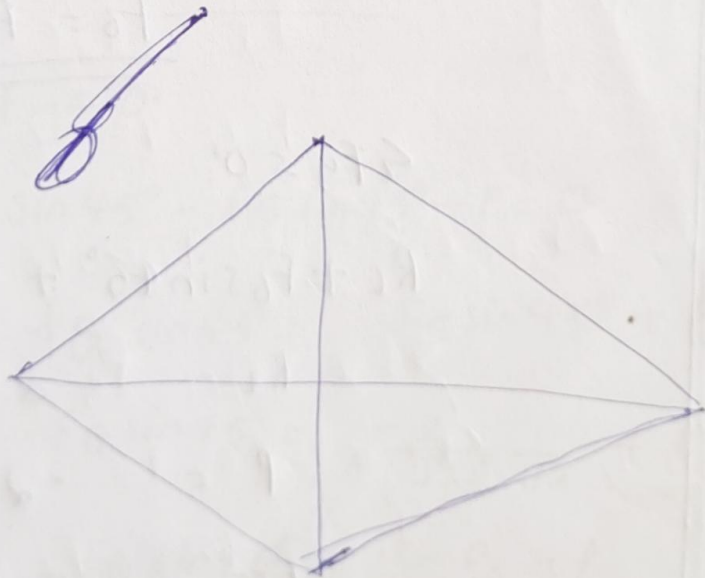
$$R_C = P \rightarrow (2)$$

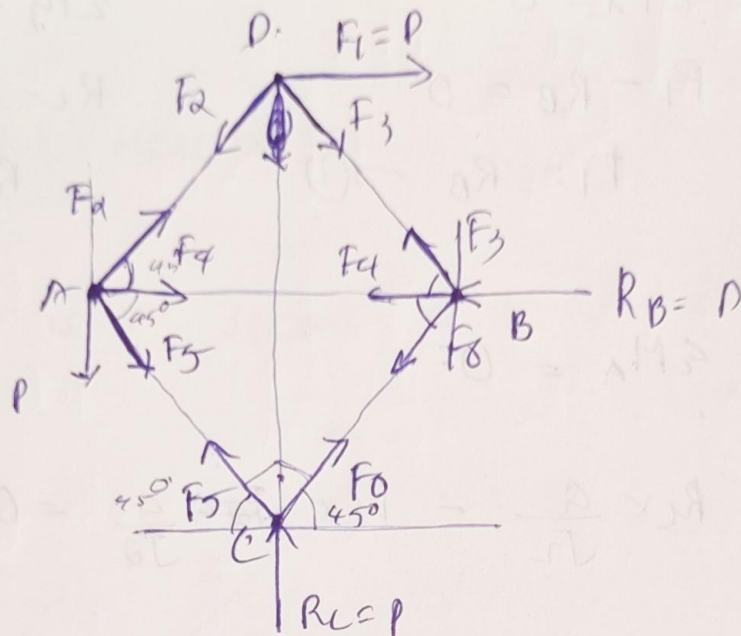
$$\sum M_A = 0$$

$$R_C \times \frac{a}{\sqrt{2}} - F_1 \times \frac{a}{\sqrt{2}} = 0$$

$$\frac{R_C \times a}{\sqrt{2}} = \frac{F_1 \times a}{\sqrt{2}}$$

$$\underline{\underline{F_1 = R_C = P = R_B}}$$





About C

$$\sum F_x = 0$$

$$F_6 \cos 45^\circ - F_5 \cos 45^\circ = 0$$

$$\underline{F_6 = F_5}$$

$$\sum F_y = 0$$

$$R_C + F_6 \sin 45^\circ + F_5 \sin 45^\circ = 0$$

$$R_C = -2 F_5 \sin 45^\circ$$

$$P = -2 F_5 \sin 45^\circ$$

$$F_5 = \frac{-P}{2 \sin 45^\circ}$$

$$F_5 = \frac{-P}{\sqrt{2}} = F_6$$

About A

$$\sum F_x = 0$$

$$F_4 \cos 45^\circ$$

$$F_4 + F_2 \cos 45^\circ + F_5 \cos 45^\circ = 0$$

$$F_4 = -F_2 \cos 45^\circ - F_5 \cos 45^\circ$$

$$F_4 = -\cos 45^\circ \left[F_2 - \frac{P}{\sqrt{2}} \right]$$

$$F_4 = \frac{-F_2}{\sqrt{2}} + \frac{P}{2} \rightarrow (3)$$

$$\sum F_y = 0$$

$$F_4 \sin 45^\circ - F_5 \sin 45^\circ - P = 0$$

$$F_4 \sin 45^\circ = F_5 \sin 45^\circ + P$$

$$F_4 \sin 45^\circ = \frac{-P}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + P$$

$$F_4 \sin 45^\circ = -\frac{P}{2} + P$$

$$F_4 \sin 45^\circ = \frac{P}{2}$$

$$F_4 = \frac{P}{2} \times \sqrt{2}$$

$$F_Q = \frac{P}{\sqrt{2}}$$

15.
In (3)

$$F_Q = \frac{-P}{\sqrt{2}} + \frac{P}{\sqrt{2}} = 0$$

Along B

$$\sum F_n = 0$$

$$-F_Q - F_3 \cos 45^\circ - F_6 \cos 45^\circ - P = 0$$

$$F_3 \cos 45^\circ = 0 - F_Q - F_6 \cos 45^\circ - P$$

$$F_3 \cos 45^\circ = 0 - \left(\frac{-P}{\sqrt{2}} \right) + \frac{P}{\sqrt{2}} - P$$

$$F_3 \cos 45^\circ = \frac{P}{2} - P$$

$$F_3 \cos 45^\circ = -\frac{P}{2}$$

$$F_3 = \frac{-P}{2} \times \sqrt{2}$$

$$= \underline{\underline{\frac{-P}{\sqrt{2}}}}$$

$$\therefore F_1 = P$$

$$F_2 = \frac{P}{\sqrt{2}}$$

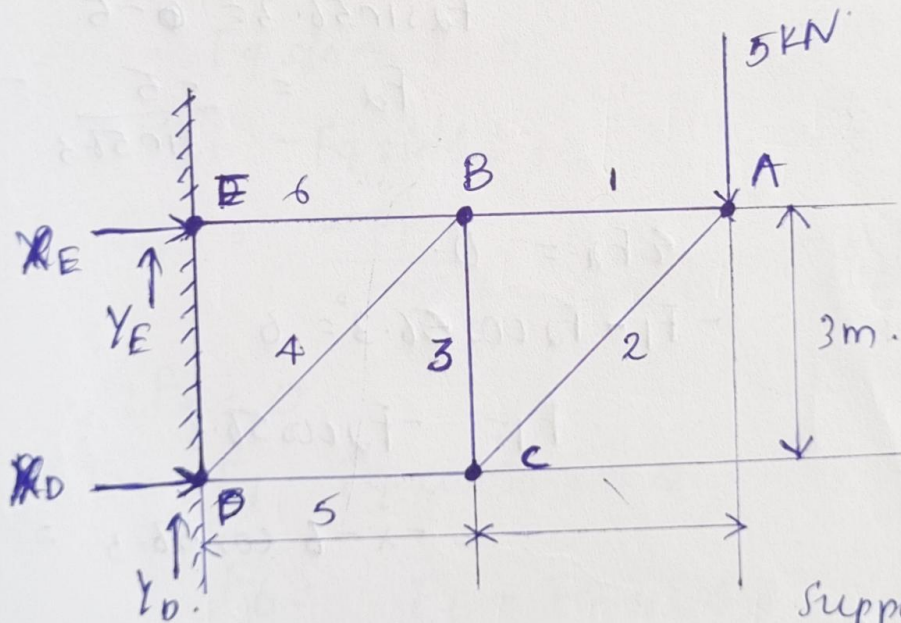
$$F_3 = \frac{-P}{\sqrt{2}}$$

$$F_4 = 0$$

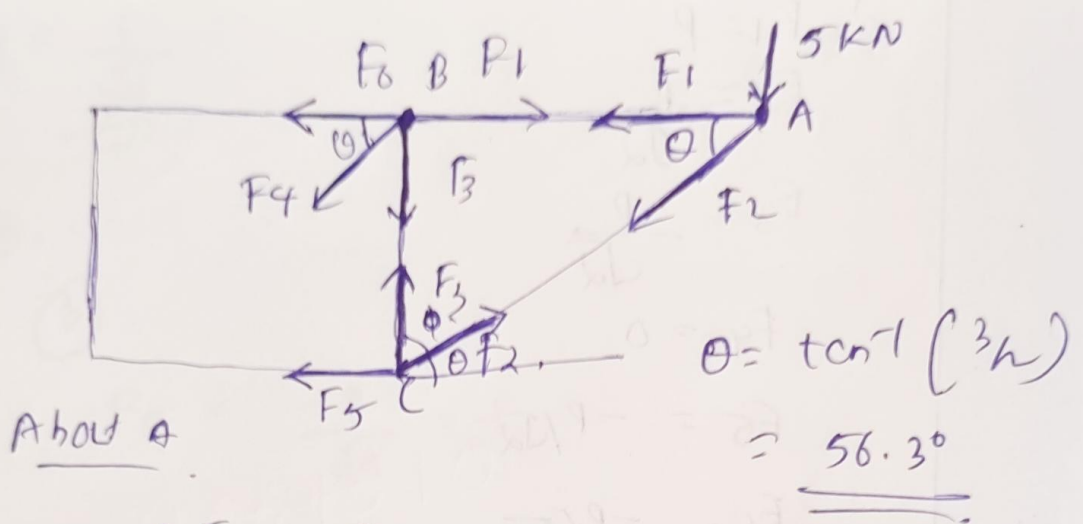
$$F_5 = -P/\sqrt{2}$$

$$F_6 = -P/\sqrt{2}$$

Q.



Support at only
one end and
not at the other
such trusses is called as the
cantilever support
truss.



About A

$$\sum F_y = 0$$

$$-5 - F_2 \sin 56.3 = 0$$

$$F_2 \sin 56.3 = -5$$

$$F_2 = \frac{-5}{\sin 56.3} = -6 \text{ kN}$$

$$\sum F_x = 0$$

$$-F_1 - F_2 \cos 56.3 = 0$$

$$F_1 = -F_2 \cos 56.3$$

$$= -(-6) \cos 56.3 = 3.32 \text{ kN}$$

About C

$$\sum F_x = 0$$

$$F_2 \sin 33.7 - F_5 = 0$$

$$F_5 = F_2 \sin 33.7$$

$$= -6 \sin 33.7$$

$$= -3.32 \text{ kN}$$

$$\sum F_y = 0$$

$$F_3 + F_2 \cos 33.7 = 0$$

$$\begin{aligned} F_3 &= -F_2 \cos 33.7 \\ &= -\cancel{6} \cos 33.7 \\ &= 4.9 \text{ kN} = \underline{\underline{5 \text{ kN}}} \end{aligned}$$

Method of
section

About B , $\theta = \underline{\underline{56.3}}$

$$\sum F_y = 0$$

$$\cancel{F_1} = \cancel{6}$$

$$-F_3 - F_4 \sin 56.3 = 0$$

$$F_3 = -F_4 \sin 56.3$$

$$F_4 = \frac{-F_3}{\sin 56.3} = \frac{-5}{\sin 56.3} = \underline{\underline{-6 \text{ kN}}}$$

$$\sum F_x = 0$$

$$-F_6 + F_1 - F_4 \cos 56.3 = 0$$

$$\begin{aligned} F_6 &= F_1 - F_4 \cos 56.3 \\ &= 3.32 + 6 \cos 56.3 \\ &= 6.64 \text{ kN} \\ &\underline{\underline{\hspace{2cm}}} \end{aligned}$$